Customer Purchasing Behaviour Analysis: SQL Project Report

Prepared by Mandela Chalingi Lotilo | Tool: SQL (MySQL) | Dataset: 3,863 customer records, 18 columns | Currency: USD | Location: USA

1. Executive Summary

This project analyses 3,863 customer purchase records from a USA retail business to answer 16 business questions on revenue performance, discount effectiveness, product performance, customer loyalty and customer segments. Total revenue across the dataset is **USD 230,629**, with an average purchase of **USD 59.70**.

The five findings that matter most:

- **Revenue is driven by customer volume, not spend per customer.** Male customers generate 67.4% of revenue (USD 155,438) against 32.6% for female customers (USD 75,191), yet female customers spend slightly more per purchase (USD 60.25 vs USD 59.44). The gap comes from the number of customers, roughly 2,615 men to 1,248 women.
- **Discounts do not lift spending.** Discounted purchases average USD 59.12, while full-price purchases average USD 60.13, a gap of only USD 1.01 (about 1.7%) in favour of full price. Of the 1,640 customers who received a discount, 816 (49.8%) still spent at or above the overall average, so those discounts may have been unnecessary.
- **Clothing and Accessories carry the business.** Together they produce 76.5% of revenue (USD 176,448 of USD 230,629).
- **The customer base is highly loyal.** 79.9% of customers (3,085) are classed as Loyal, with more than 10 previous purchases, and only 2.1% are new.
- **Spending per customer is remarkably uniform.** Average spend varies by less than USD 2.20 across shipping types, no more than USD 1.80 across age bands and less than USD 0.90 across gender and subscription groups. Growth is more likely to come from reaching more customers than from changing how much each one spends.

2. Project Overview

2.1 Objective

To use SQL to turn raw customer transaction data into clear, decision-ready insights for a retail business, focusing on five areas:

- Revenue performance
- Discount effectiveness
- Product performance
- Customer loyalty and segments
- Revenue by age

2.2 Dataset

Item	Detail
Table	customers
Rows	3,863
Columns	18
Currency	USD
Location	USA
Key fields used	customer_id, age, gender, item_purchased, category, purchase_amount, location, season, review_rating, subscription_status, shipping_type, discount_applied, previous_purchases, payment_method, frequency_of_purchases

2.3 Data Preparation

The column `Customer ID` was renamed to `customer_id` using `ALTER TABLE ... RENAME COLUMN` so that it could be referenced without backticks in later queries. The table was inspected with `SHOW TABLES`, `DESCRIBE` and `SELECT *` before analysis began.

Totals reconcile across the analysis. Revenue by gender, category, season, subscription status and age band each add up to the same USD 230,629, and the loyalty segments (3,085 + 695 + 83) add up to all 3,863 customers. This confirms no records were lost in

grouping.

3. Methodology

All analysis was done in SQL. The script uses these techniques:

Technique	Used in
Aggregation (`SUM`, `AVG`, `COUNT`) with `GROUP BY`	Questions 1, 3, 4, 6, 11, 12, 13, 14, 16
Subquery for a benchmark value	Question 2 (overall average purchase)
Conditional aggregation with `CASE WHEN`	Questions 5, 7
Common Table Expressions (CTEs)	Questions 8, 9, 10
Window functions (`RANK() OVER (PARTITION BY ...)`, `SUM() OVER`)	Questions 9, 15
Customer segmentation and age banding with `CASE`	Questions 8, 10
`ORDER BY` and `LIMIT` for ranking	Questions 4, 5, 16

Definitions used in this report

- *Repeat buyer* (Question 7): a customer with more than 5 previous purchases.
- *Loyalty segments* (Question 10): New = 1 previous purchase; Returning = 2 to 10; Loyal = more than 10.
- *Age bands* (Question 8): 18-25, 26-35, 36-45, 46-55, 56-65 and 66+.
- Percentages and customer counts marked "derived" were calculated from the script's outputs and not queried directly.

4. Key Findings

4.1 Revenue Performance

Revenue by gender (Questions 1 and 11)

Gender	Customers (derived)	Revenue	Share of revenue	Average spend
Male	about 2,615	USD 155,438	67.4%	USD 59.44
Female	about 1,248	USD 75,191	32.6%	USD 60.25

Female customers spend about 1.4% more per purchase, but there are roughly half as many of them. The revenue gap is a customer-acquisition gap, not a spending gap.

Revenue by category (Question 12)

Category	Revenue	Share
Clothing	USD 102,968	44.6%
Accessories	USD 73,480	31.9%
Footwear	USD 35,778	15.5%
Outerwear	USD 18,403	8.0%

Clothing is by far the largest single contributor, at USD 102,968 against USD 127,661 for the other three categories combined. Outerwear is the weakest category and has only two products.

Revenue by season (Question 13)

Season	Revenue	Share
Fall	USD 59,620	25.8%
Winter	USD 58,035	25.2%
Spring	USD 57,877	25.1%
Summer	USD 55,097	23.9%

Revenue is spread almost evenly across the year. Fall leads Summer by USD 4,523 (about 8.2%), so seasonality exists but is mild. Stock planning does not need to swing sharply between seasons.

Revenue by age group (Question 8)

Age band	Customers (derived)	Revenue	Share	Average spend
46-55	about 743	USD 44,927	19.5%	USD 60.47
26-35	about 736	USD 44,008	19.1%	USD 59.79
56-65	about 745	USD 43,930	19.0%	USD 58.97
36-45	about 722	USD 42,853	18.6%	USD 59.35
18-25	about 566	USD 34,281	14.9%	USD 60.57
66+	about 351	USD 20,630	8.9%	USD 58.77

Four bands between ages 26 and 65 each contribute 18.6% to 19.5% of revenue. The 18-25 band has the highest average spend (USD 60.57) but contributes less because it has fewer customers. Note that this band spans 8 years against 10 for the middle bands, which partly explains the smaller count. The 66+ band is the smallest at 8.9% of revenue.

Revenue by location (Question 16)

Rank	State	Revenue
1	Montana	USD 5,717
2	California	USD 5,605
3	Illinois	USD 5,537
4	Idaho	USD 5,438
5	Nevada	USD 5,379

The top five states together bring in USD 27,676, or 12.0% of revenue, and the gap between first and fifth is only USD 338. No state dominates, so revenue is geographically diversified.

4.2 Discount Effectiveness

Who receives discounts (Question 2)

- 1,640 customers (42.5% of all customers) received a discount.
- 816 of them (49.8%) still spent at or above the overall average of USD 59.70. These customers make up 21.1% of all customers.

Do discounts increase spending? (Question 14)

Discount applied	Orders	Average spend
Yes	1,640	USD 59.12
No	2,223	USD 60.13

Full-price customers spend USD 1.01 more on average. Discounts are not producing larger baskets, and a large share of discounted customers were already willing to spend above average without one.

Products most dependent on discounts (Question 5)

Product	Share of purchases made with a discount
Hat	49.67%
Sneakers	49.66%
Coat	48.43%
Sweater	47.20%
Pants	46.43%

For these products nearly half of all sales involve a discount, so a discount may be the expected price rather than a special offer.

4.3 Product Performance

Top-rated products (Question 4)

Product	Average rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

The top five are separated by only 0.07 rating points, so customer satisfaction is consistent across products and no product is a clear outlier.

Best-selling products within each category (Question 9)

Category	1st	2nd	3rd
Accessories	Jewelry (170)	Sunglasses and Belt (159 each, tied)	n/a
Clothing	Blouse (169)	Shirt and Pants (168 each, tied)	n/a
Footwear	Sandals (159)	Shoes (147)	Sneakers (145)
Outerwear	Jacket (163)	Coat (159)	n/a

Order counts within each category are very close. Because `RANK()` gives tied items the same position, Sunglasses and Belt share second place in Accessories, and Shirt and Pants share second place in Clothing.

Some products appear in more than one list. Sandals are both the best-selling Footwear item and the second-highest rated product overall. Hats and Coats appear among the most discount-dependent products. Hats are also among the top-rated, so heavy discounting is not a sign of poor quality, though the data cannot show how much of their volume depends on it.

4.4 Customer Loyalty and Subscription

Loyalty segments (Question 10)

Segment	Definition	Customers	Share
Loyal	More than 10 purchases	3,085	79.9%
Returning	2 to 10 purchases	695	18.0%
New	1 purchase	83	2.1%

Almost four in five customers are Loyal. This is a strength, but the small New segment also suggests the business is acquiring few new customers.

Subscription impact (Questions 6 and 7)

Status	Customers	Average spend	Total revenue	Repeat-buyer rate
Not subscribed	2,830	USD 59.81	USD 169,252	88.4% (2,503)
Subscribed	1,033	USD 59.42	USD 61,377	90.9% (939)

- Non-subscribers produce more total revenue (73.4%) only because there are almost three times as many of them.
- Subscribers do not spend more per purchase (USD 0.39 less on average).
- Subscribers are slightly more likely to be repeat buyers, by 2.5 percentage points. The value of subscription appears to lie in retention, not in higher spend.

4.5 Shipping and Payment

Average purchase by shipping type (Question 3)

Shipping type	Average purchase
2-Day	USD 60.59
Free Shipping	USD 60.37
Express	USD 60.34
Store Pickup	USD 59.91
Next Day Air	USD 58.62
Standard	USD 58.41

The spread from highest to lowest is USD 2.18 (about 3.7%). Shipping choice has only a small association with spend.

Payment method and purchase frequency (Question 15)

- PayPal is the most used payment method, with 673 customers (17.4% of all customers).
- The largest single group is Credit Card customers who buy every 3 months, with 112 customers (2.9%).
- Customers are spread thinly across payment methods and frequencies, so no single combination dominates.

5. Recommendations

1. **Review the discount strategy.** Discounts are not raising spend, and about half of discounted customers already spend above average. Test targeting discounts at lower-spending or new customers, and run a controlled comparison before cutting them entirely.
2. **Reconsider discount-heavy products.** Hat, Sneakers, Coat, Sweater and Pants are discounted on 46% to 50% of sales. Trial full-price periods for one or two of them to see whether volume holds.
3. **Grow the female customer base.** Female customers spend slightly more per purchase, so reaching more of them is the clearest revenue lever. Marketing could be tested toward women in the categories they already buy.
4. **Protect and promote Clothing and Accessories.** They deliver 76.5% of revenue. Keep these ranges well stocked and use them to anchor promotions.
5. **Treat subscription as a retention tool.** Subscribers repeat-buy more often but do not spend more. Judge the programme on retention and lifetime value, not basket size.
6. **Invest in acquisition.** With only 2.1% of customers new, the business relies heavily on existing buyers. A modest acquisition campaign would reduce that dependence.
7. **Target younger and older age groups differently.** The 18-25 band spends the most per purchase but is under-represented, so there is room to grow it. The 66+ band is the smallest and may need different products or channels.
8. **Keep stock planning flexible.** With seasonal revenue within 8.2% of each other, plan for a modest Fall and Winter lift without heavy seasonal stockpiling.

6. Limitations

- **Observational data.** The results show associations, not cause and effect. For example, the data cannot say whether discounts brought in customers who would not otherwise have bought.
- **Discount effect is measured on spend only.** Purchase frequency and conversion are not captured by the average-spend comparison.
- **Very small differences.** Many gaps (shipping, rating, age spend, subscription spend) are small, and no significance testing was done, so they should not be over-interpreted.
- **Unequal age bands.** The 18-25 band covers 8 years and 66+ is open-ended, which affects band comparisons.
- **Single-period snapshot.** There is no date field in the analysis, so trends over time cannot be assessed.
- **"Repeat buyer" threshold.** The definition of more than 5 previous purchases is a choice, and results would change with a different cut-off.

7. Next Steps

- Build a Power BI dashboard (or extend the existing one) to present these findings visually.
- Add significance testing in Python to check whether small differences, such as discount vs full price, are statistically meaningful.
- Calculate customer lifetime value by segment and subscription status.
- Explore purchase frequency and payment method against revenue to find high-value habits.
- Build a simple customer clustering model (for example RFM-style segmentation) to go beyond the age and loyalty bands.

8. Appendix: Script Quality Notes

Small changes would make the SQL script cleaner for portfolio review:

- **Fix typos in comments**, for example "Smummer", "forth", "fift", "postion", "purcase", "freguency", "puechases", "buy age group" (should be "by"), "average send" (should be "spend") and "dicount".
- **Standardise comment style.** Some answers use "leads/returns/takes" while others use plain sentences. Keep each finding to a short consistent pattern.

- **Handle NULLs in Question 10.** The `CASE` has no `ELSE`, so a customer with 0 previous purchases would return NULL. The segment counts currently add up to 3,863, so none exist, but adding `ELSE 'Unclassified'` makes the query safer.
- **Add percentage columns to the queries** (for example share of total revenue) so the findings do not rely on manual calculation.
- **Rename Question 2's heading** to something like "Discounted customers who still spent above average" so it matches what the query does.
- **Comment out the one-time `ALTER TABLE`** after running it so the script can be re-run without error.